Exploratory Data Analysis (EDA) and Outlier Detection Report

This report presents the findings from the Exploratory Data Analysis (EDA) and statistical analysis performed on the provided cleaned_dataset.csv. The analysis focuses on understanding data features, identifying trends in customer behavior and demand, and detecting potential outliers

1. Data Inspection and Descriptive Statistics

The dataset contains 3,271 records and 13 features. Crucially, there are no missing values in the dataset, indicating a high level of data quality or successful prior cleaning.

A. Numerical Feature Summary

The descriptive statistics reveal that the features MaxBookees, Number Booked, and PriceINR have been scaled or standardized as evidenced by their mean values close to zero, standard deviations close to one, and the presence of negative values. This is an important consideration for model building.

	count	mean	std	min	25%	50%	75%	max
MaxBookees	3271	32.2244	15.1214	2	24	30	35	70
Number Booked	3271	17.126	9.51535	1	10	16	24	64
PriceINR	3271	1852.71	792.637	499	1299	1499	2499	3999
Hour	3271	13.3702	3.92096	6	10	12	17	20
Month	3271	5.01345	0.816386	4	4	5	6	6
Week	3271	20.0061	3.75935	13	17	20	23	26
DayOfWeekNum	3271	2.41608	1.73582	0	1	2	4	6
occupancy_rate	3271	0.572207	0.319093	0.02	0.34	0.56	0.8	5.17

The occupancy_rate (calculated as Number Booked / MaxBookees in the original scale) has a mean of approximately 57.2%, suggesting a generally healthy demand level.

B. Categorical Feature Summary

The analysis of categorical features highlights the distribution across different sites, activities, and days of the week.

ActivitySiteID	count
HXP	771
SBP	731
BRP	727
NBL	529
TSC	513

DayOfWeek	count
Wednesday	677
Tuesday	581
Thursday	540
Monday	539
Friday	495
Saturday	266
Sunday	173

Key Observations:

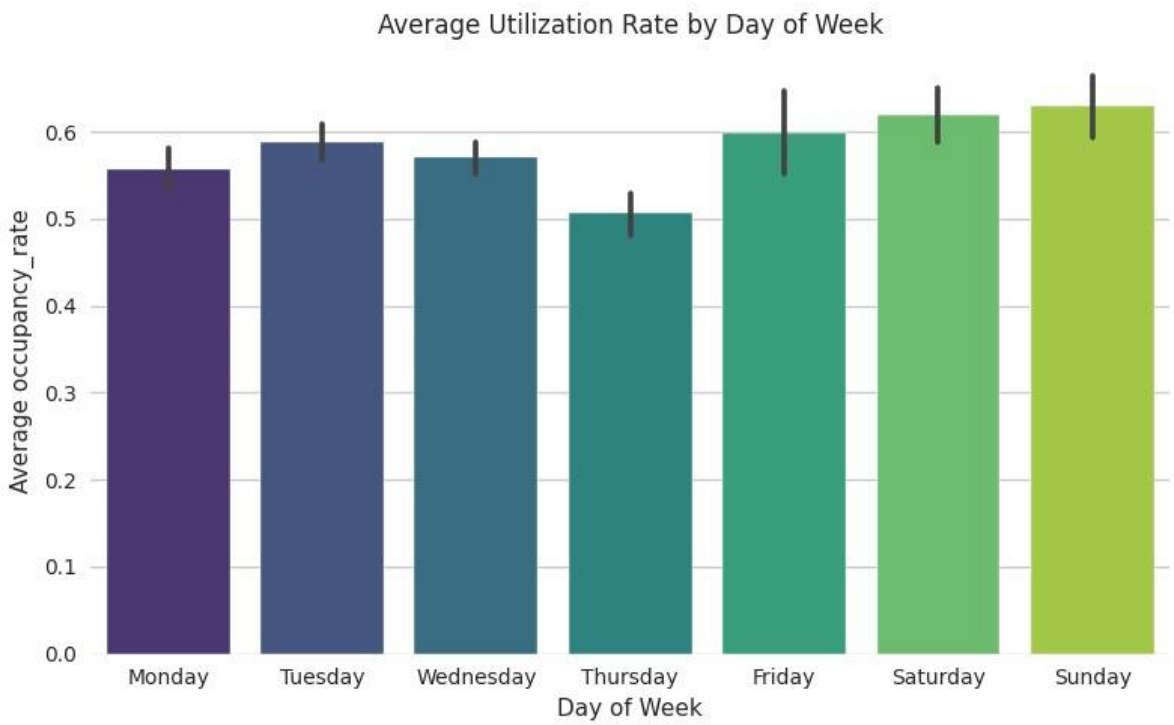
•Activity Sites: The activity is distributed across five main sites (HXP, SBP, BRP, NBL, TSC) with relatively even distribution among the top three.

•Day of Week: Demand is highest mid-week, with Wednesday having the highest number of bookings, followed by Tuesday and Thursday. Weekends (Saturday and Sunday) show significantly lower activity counts, which could be due to number of lower demand in bookings.

2. Demand Pattern Analysis (Visualizations)

Visualizations were generated to reveal booking trends and seasonal patterns.

A. Demand by Day of Week

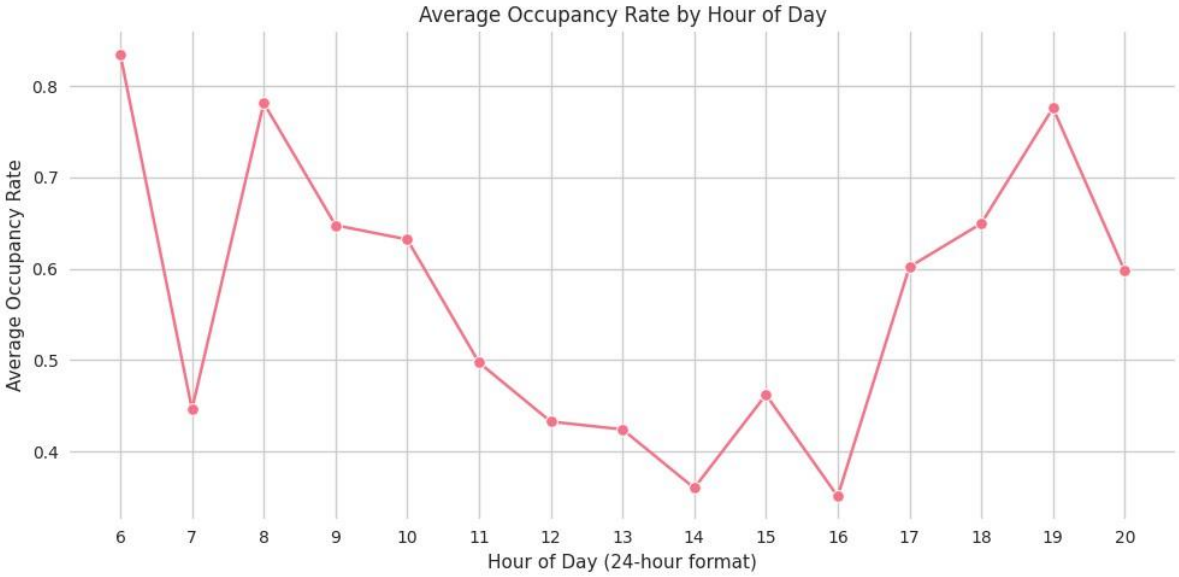


Based on above plot:

- **Highest Utilization**: **Sunday** shows the highest average utilization rate, indicating that activities scheduled on this day are, on average, the most occupied.
- **Lowest Utilization**: **Monday** and **Thursday** appear to have the lowest average utilization rates, suggesting that activities on these days might experience lower occupancy compared to other days.
- **Mid-Range Days**: Other weekdays like Tuesday, Wednesday, and Friday fall somewhere in between, with moderate utilization rates.

This analysis helps in understanding the weekly demand patterns, highlighting both peak and low utilization days for class scheduling and resource management.

B. Demand by Hour of Day



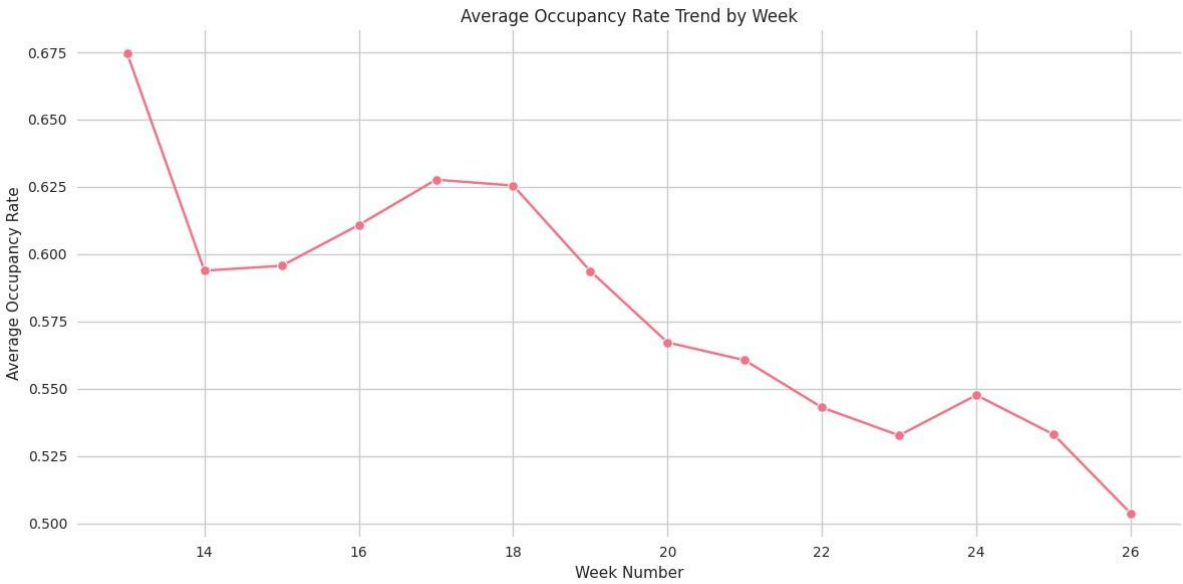
The 'Average Occupancy Rate by Hour of Day' plot, generated from the hourly demand data, reveals the following trends:

- **Morning Peaks**: There's a notable high average occupancy rate around **6 AM** and then again around **8 AM**, suggesting early morning activities are quite popular.
- **Mid-Day Dip**: Occupancy rates tend to decrease during mid-day hours, reaching their lowest point around **2 PM** to **4 PM**.
- **Evening Surge**: There's a significant increase in occupancy rates in the evening, with a peak around **7 PM**, indicating high demand for activities after typical work hours. In summary, the plot shows activity throughout the day, with strong demand in the early mornings and evenings, and lower demand in the afternoons.

C. Seasonal Trend

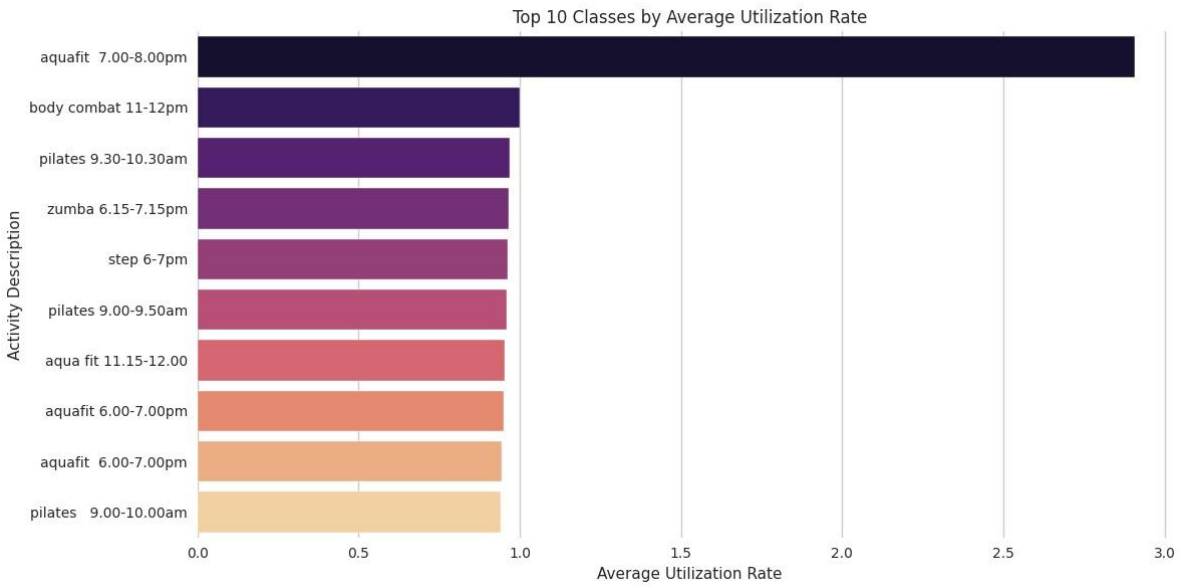
- **General Decline**: The plot shows a general decreasing trend in the average occupancy rate over the observed weeks (from week 13 to week 26). This suggests that as time progresses through this period, the overall utilization of activities tends to go down.

- **Fluctuations**: While there's a general downward trend, there are minor fluctuations. For instance, there might be slight increases or plateaus in certain weeks before continuing the decline.



C. High-Demand Classes: -

The top 10 classes by average occupancy rate (Figure 4) indicate which activities are most popular relative to their capacity. This information is vital for dynamic pricing, as these classes may tolerate higher price points.

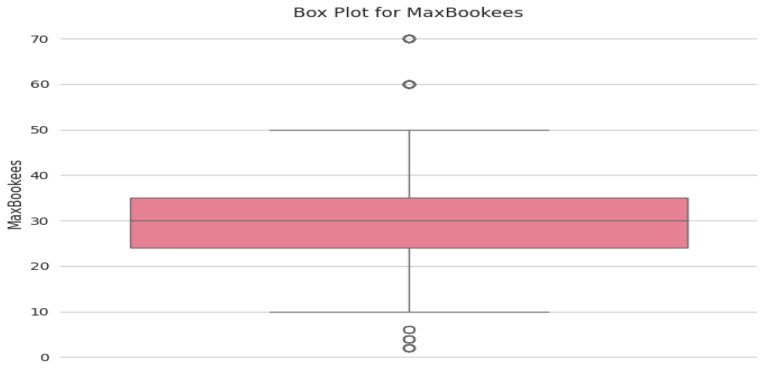


- **Aquafit classes** appear very frequently among the top utilized classes, with 'aquafit 7.00-8.00pm' having a significantly high utilization rate.
- **Body Combat and Pilates classes** also show high average utilization, indicating consistent demand and efficient booking for these types of activities.
- **Zumba and Step classes** are also present in the top 10, demonstrating their popularity and high occupancy

This analysis is crucial for understanding which specific activities are most efficiently utilizing their available slots, potentially guiding decisions on class scheduling, resource allocation, and marketing efforts.

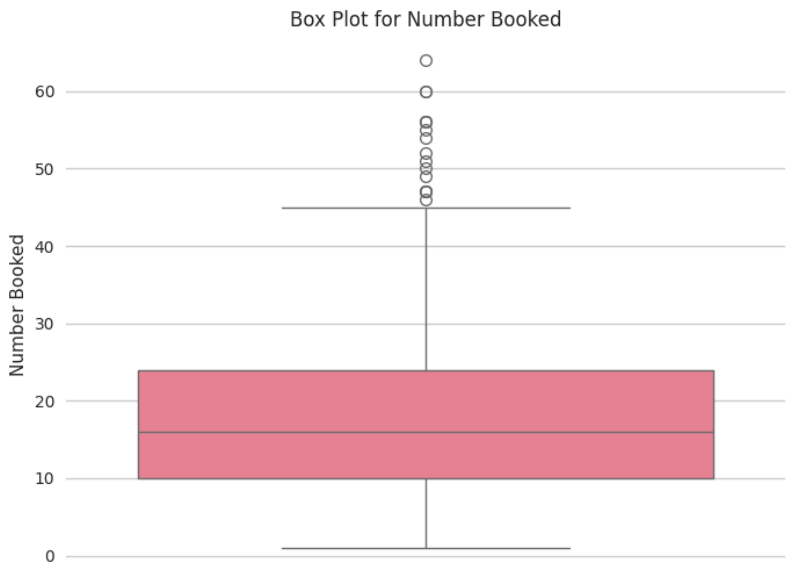
3. Outlier and Anomaly Identification

1. **Box Plot for MaxBookees :** This plot visualizes the distribution of the MaxBookees (maximum booking capacity) for each activity. The outlier analysis showed that 16.81% of the data points (550 entries) are identified as outliers for MaxBookees. This indicates that there's a significant number of classes with capacities that are either much higher or much lower than the typical range of class sizes.



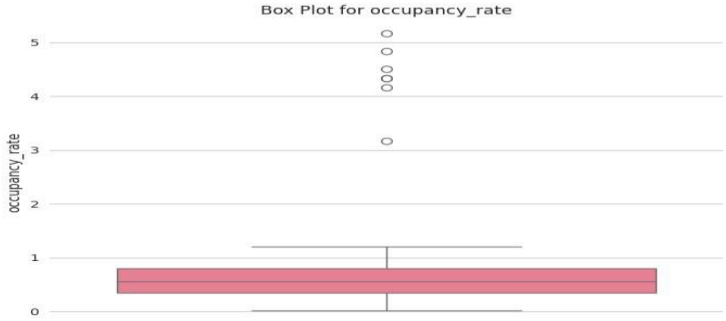
2. Box Plot for Number Booked

This plot displays the distribution of the Number Booked (actual number of people who booked a class). For Number Booked the outlier analysis detected 0.67% of the data points (22 entries) as outliers. This suggests that while most classes have a number booked count within the expected range there are a few instances where the actual bookings were unusually high or low compared to the majority of classes



3. Occupancy Rate

This plot displays the distribution of the occupancy rate for activities. For occupancy rate, the outlier analysis detected 0.21% of the data points (7 entries) as outliers. This suggests that while most activities have an occupancy rate within the typical range, there are a few rare instances where the occupancy rate was either exceptionally high or unusually low compared to the majority of classes.



4. PriceINR:

   The PriceINR feature shows no outliers based on the IQR method, suggesting a consistent and controlled range of prices across the dataset.

4. Conclusion and Next Steps

The Exploratory Data Analysis has successfully characterized the dataset and revealed several key insights for dynamic pricing:

a. Demand Drivers: Demand is highest week-end (Sunday) and during morning (**6 AM** and **8 AM**) and evening (7 PM) hours.
b. Seasonal Effect: A noticeable decline in average occupancy rate is observed from Week 13 to Week 26, suggesting a need to adjust pricing or promotions seasonally.
c. High-Value Classes: Specific classes with consistently high occupancy rates have been identified, which are prime candidates for price optimization.
d. Data Quality Concern: The significant outliers in MaxBookees require further investigation.